

CampusConnect

A Smart Campus Exchange Platform

Requirements Specification Document

(Business Requirements Document — BRD)

Prepared for: Requirements Specification Assignment

Course Instructor: JayPrakash Lalchandani

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Team Members

Dev Barot (202512069) — Project Manager & System Architect

Dhrumil Doshi (202512088) — Authentication & User Management Lead

Nishka Shah (202512035) — Marketplace & Listing Management Lead

Heer Mehta (202512095) — AI & Intelligent Services Lead

Heer Shah (202512024) — Search, Chat & Notification Lead

Charmi Bhayani (202512028) — Rental & Transaction Management Lead

Kajal Varlani (202512017) — Admin Panel & Analytics Lead

Deep Boghara (202512076) — Database, DevOps & Quality Assurance Lead

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements, user stories, use cases, interface requirements, initial UI/UX wireframes, and acceptance criteria for CampusConnect, a smart campus exchange platform. It translates the goals and scope established in the Project Proposal, and the execution plan established in the Project Plan, into a testable requirements baseline that the development team can build and validate against.

1.2 Scope

CampusConnect is a web-based marketplace exclusive to verified university students, supporting buying, selling, renting, exchanging, and giving away items. The platform integrates AI-assisted listing creation, semantic matching between wanted posts and listings, OCR-based duplicate detection for notes and question papers, real-time chat and notifications, QR-verified rental handovers, and an administrative moderation and analytics layer. This document covers requirements for the student-facing web client, the administrator dashboard, and the supporting backend, AI, and real-time services described in the Technology Stack.

1.3 Referenced Documents

- CampusConnect — Project Proposal (scope, technology stack, roles and responsibilities).
- CampusConnect — Project Plan (team timeline, module ownership, individual execution plans, dependencies, and task board).

1.4 Definitions and Acronyms

Term	Definition
BRD	Business Requirements Document
FR / NFR	Functional Requirement / Non-Functional Requirement
RBAC	Role-Based Access Control
JWT	JSON Web Token, used for session authentication
OCR	Optical Character Recognition
PYQ	Previous Year (Question) Papers
CRUD	Create, Read, Update, Delete
Wanted Post	A listing created by a student describing an item they are looking for
Match Event	A system-generated event raised when a new listing satisfies an open wanted post
QR Handover	The QR-code-based verification step used to confirm a physical item exchange

2. Functional Requirements

Functional requirements are grouped by platform module, matching the ownership structure defined in the Project Plan. Each requirement is uniquely numbered for traceability into user stories, wireframes, and acceptance criteria.

2.1 Authentication & User Management

ID	Requirement
FR-1.1	The system shall allow account registration only with a valid, verified university email address.
FR-1.2	The system shall authenticate users using JWT-based sessions with bcrypt-hashed passwords.
FR-1.3	The system shall enforce role-based access control distinguishing Student and Administrator permissions.
FR-1.4	Users shall be able to view and edit their profile, including name, program/year, and contact preferences.
FR-1.5	The system shall maintain a ratings and reputation score for each user, updated after completed transactions.
FR-1.6	Users shall be able to store payment methods and complete payments through an integrated payment gateway.
FR-1.7	The system shall generate separate buyer and seller confirmations for each completed item receipt.
FR-1.8	The system shall provide a personal sustainability dashboard summarizing money saved, items reused, and estimated CO ₂ impact.

2.2 Marketplace & Listing Management

ID	Requirement
FR-2.1	Users shall be able to create, edit, and delete listings with title, description, category, condition, images, price, and listing type.
FR-2.2	The system shall support administrator-configurable item categories.
FR-2.3	The system shall support upload and storage of one or more images per listing via cloud storage.
FR-2.4	Users shall be able to browse and search listings using category, price range, condition, distance, and listing-type filters.
FR-2.5	The system shall offer AI-assisted listing creation that pre-fills category, condition, and price from an uploaded photo.
FR-2.6	The system shall support a clean transition of a listing into a rental workflow when the listing type is Rent.
FR-2.7	The system shall fully support the buy, sell, rent, exchange, and giveaway transaction workflows end to end.

2.3 AI & Intelligent Services

ID	Requirement
FR-3.1	The AI service shall generate listing suggestions (title, category, condition, price) from an uploaded image.
FR-3.2	The AI service shall estimate a fair market price for a listing based on comparable items.
FR-3.3	The system shall run semantic matching between open wanted posts and newly created listings and raise match events.
FR-3.4	The system shall perform OCR-based duplicate detection on uploaded notes/PYQ documents before publishing.
FR-3.5	The AI service shall expose versioned FastAPI endpoints and support mocked responses for independent module development.

2.4 Search, Chat & Notifications

ID	Requirement
FR-4.1	Users shall be able to perform keyword search across active listings with sorting and filtering.
FR-4.2	Users shall be able to save searches and be notified when new matching listings are published.
FR-4.3	Users shall be able to maintain a wishlist of items of interest.
FR-4.4	The system shall provide real-time in-app chat between a listing's buyer and seller using Socket.io.
FR-4.5	The system shall deliver real-time notifications for matches, new messages, rental due dates, and moderation actions.
FR-4.6	The notification pipeline shall be extensible to email/push delivery for users who are not actively online.

2.5 Rental & Transaction Management

ID	Requirement
FR-5.1	Users shall be able to submit a rental request specifying the desired rental period.
FR-5.2	The system shall track transaction state through requested, approved, active, completed, and cancelled/disputed.
FR-5.3	The system shall generate a QR code for item handover and verify it at pickup and return.
FR-5.4	The system shall automatically track due dates and trigger rental reminder notifications ahead of the due date.
FR-5.5	Users shall be able to view a rental dashboard listing active rentals, pending requests, and rental history.

2.6 Admin Panel & Analytics

ID	Requirement
FR-6.1	Administrators shall be able to approve or reject submitted listings prior to publication.
FR-6.2	Administrators shall be able to manage user accounts, including suspension and verification actions.

ID	Requirement
FR-6.3	Administrators shall be able to create, edit, and remove marketplace categories.
FR-6.4	Administrators shall be able to view and resolve user-submitted disputes and reports through a dedicated workflow.
FR-6.5	The admin dashboard shall present analytics on demand trends, active users, and listing volume.
FR-6.6	The system shall send moderation notifications to users affected by an administrative action.

2.7 Platform & Cross-Cutting Requirements

ID	Requirement
FR-7.1	The system shall maintain audit logs capturing key user and administrator actions.
FR-7.2	The system shall implement centralized, consistent error-handling and logging across all services.
FR-7.3	The system shall implement API rate-limiting and request throttling to protect against abuse.
FR-7.4	Each major service shall expose a health-check / uptime-monitoring endpoint.
FR-7.5	The system shall manage environment configuration and secrets outside of source control.
FR-7.6	The system shall be deployable through a CI/CD pipeline with automated build and test checks.

3. Non-Functional Requirements

Non-functional requirements define the quality attributes the platform must satisfy across all modules.

Category	ID	Requirement
Performance	NFR-1.1	Core pages (browse, listing detail) shall load within 2 seconds on a typical campus network connection.
Performance	NFR-1.2	Search and filter results shall return within 1 second for a catalog of up to 10,000 active listings.
Performance	NFR-1.3	Chat messages shall be delivered to an online recipient within 500 milliseconds under normal load.
Scalability	NFR-2.1	The backend and AI services shall be independently horizontally scalable to accommodate growth to full university enrollment.
Scalability	NFR-2.2	The messaging/event layer (Redis) shall decouple producers and consumers so individual services can scale independently.
Security	NFR-3.1	All client-server communication shall be encrypted using HTTPS/TLS.
Security	NFR-3.2	Passwords shall never be stored in plaintext; bcrypt hashing shall be used for all credentials.
Security	NFR-3.3	All API inputs shall be validated and sanitized to prevent injection and cross-site scripting attacks.
Security	NFR-3.4	Only users with a verified university email domain shall be able to access marketplace features.
Security	NFR-3.5	Sensitive configuration (API keys, database credentials) shall be stored in environment variables or a secrets manager, never in source control.
Availability	NFR-4.1	The platform shall target 99% uptime during the academic semester, excluding scheduled maintenance.
Availability	NFR-4.2	Health-check endpoints shall allow automated monitoring to detect and alert on service degradation.
Usability	NFR-5.1	The web client shall be responsive and usable on desktop, tablet, and mobile screen sizes.
Usability	NFR-5.2	A student shall be able to publish a basic listing in 3 steps or fewer when using AI-assisted creation.
Reliability	NFR-6.1	If the AI service is unavailable, listing creation shall gracefully fall back to manual entry without blocking the user.
Reliability	NFR-6.2	Transaction and payment-related data shall remain consistent even if a dependent service (e.g., notifications) fails.
Maintainability	NFR-7.1	The backend shall expose REST APIs documented with an OpenAPI specification shared across module teams.
Maintainability	NFR-7.2	Each module shall include automated unit tests integrated into the CI pipeline.

Category	ID	Requirement
Compatibility	NFR-8.1	The web client shall support the latest two major versions of Chrome, Firefox, Edge, and Safari.
Privacy	NFR-9.1	Administrative actions on user data shall be recorded in an audit log for accountability.

4. User Stories and Use Cases

4.1 Use Case Diagrams

Use cases are split by actor for readability. The Student diagram shows the primary student-facing use cases, with the AI/Matching Service and Notification Service shown as supporting actors linked only to the specific system-level use cases they participate in (dashed <<include>>/<<extend>> relationships). The Administrator diagram shows the moderation and analytics use cases separately, since administrators interact with a distinct part of the system.

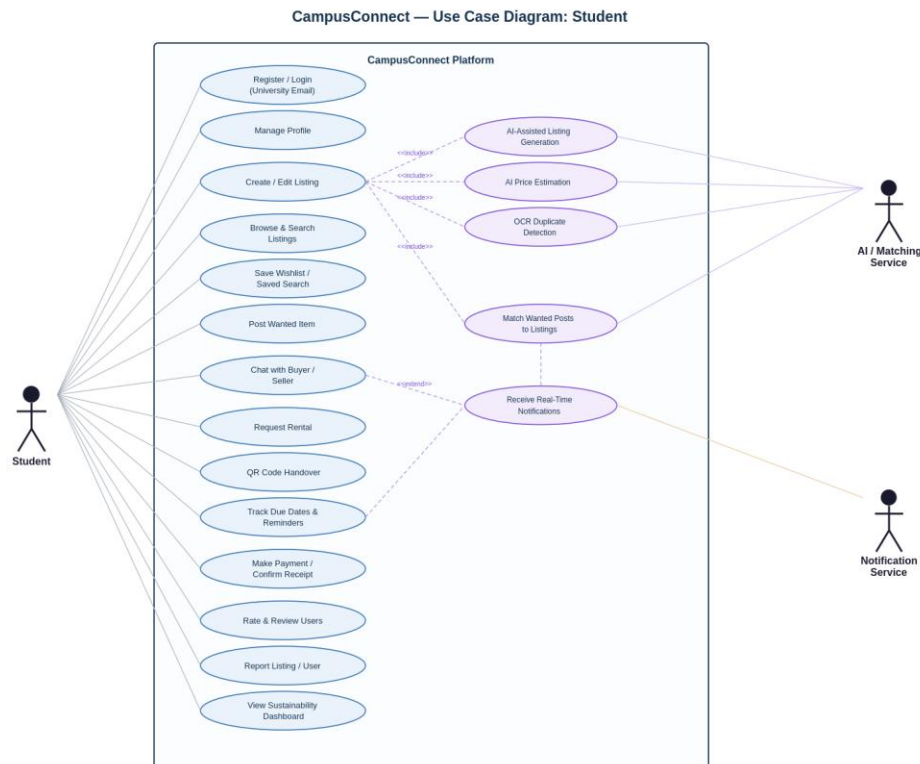


Figure 1a: CampusConnect use case diagram — Student.

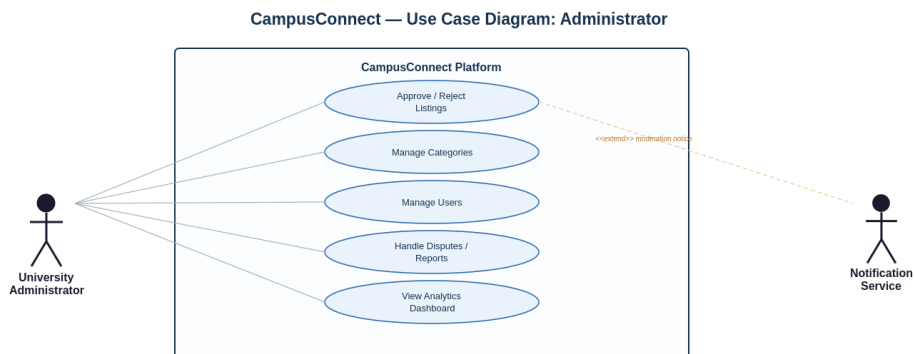


Figure 1b: CampusConnect use case diagram — Administrator.

4.2 Detailed User Stories

User stories are written from the perspective of the primary actor and mapped to the functional requirement(s) they satisfy. Priority follows MoSCoW-style classification (High = Must-have for MVP, Medium = Should-have, Low = Nice-to-have).

ID	User Story	Related FR	Priority
US-01	As a student, I want to register using my university email so that only verified students can access the marketplace.	FR-1.1	High
US-02	As a student, I want to log in securely so that my account and transaction history stay protected.	FR-1.2	High
US-03	As a student, I want to upload a photo of an item and have AI suggest the category, condition, and price so that I can list an item quickly.	FR-2.5, FR-3.1, FR-3.2	High
US-04	As a student, I want to browse and filter listings by category, price, and condition so that I can find relevant items fast.	FR-2.4	High
US-05	As a student, I want to post a “wanted” item so that I am notified automatically when someone lists a matching item.	FR-3.3	Medium
US-06	As a student, I want to chat with a buyer or seller in real time so that I can coordinate a transaction without leaving the platform.	FR-4.4	High
US-07	As a student, I want to request a rental with a specific due date so that I can borrow an item for a defined period.	FR-5.1	High
US-08	As a student, I want to confirm a handover using a QR code so that both parties have secure proof the item exchanged hands.	FR-5.3	High
US-09	As a student, I want to receive a reminder before my rental is due so that I return items on time.	FR-5.4	Medium
US-10	As a student, I want to upload notes for sale and have the system flag duplicates so that I don’t accidentally repost existing content.	FR-3.4	Medium
US-11	As a student, I want to rate a user after a transaction so that the community can trust each other’s reputations.	FR-1.5	Medium
US-12	As a student, I want to view my sustainability dashboard so that I can see the money and environmental impact I’ve saved.	FR-1.8	Low
US-13	As a student, I want to save a search so that I’m notified when new items matching my criteria are listed.	FR-4.2	Medium
US-14	As a student, I want to report an inappropriate listing or user so that the community stays safe.	FR-6.4	Medium
US-15	As an administrator, I want to review and approve new listings so that inappropriate or fraudulent content doesn’t reach students.	FR-6.1	High
US-16	As an administrator, I want to manage marketplace categories so that listings stay organized as the catalog grows.	FR-6.3	Medium

ID	User Story	Related FR	Priority
US-17	As an administrator, I want to view analytics on demand and listing trends so that I can understand marketplace health.	FR-6.5	Medium
US-18	As an administrator, I want to resolve disputes raised between students so that conflicts are handled fairly and consistently.	FR-6.4	High
US-19	As a student, I want to make a payment and separately confirm item receipt so that funds aren't released until I've verified the item.	FR-1.6, FR-1.7	High
US-20	As a platform operator, I want an audit log of key actions so that any dispute or incident can be investigated.	FR-7.1	Medium

5. Interface Requirements

5.1 User Interfaces

The primary user interface is a responsive React web application shared by students and administrators, using role-based views. Key screens include: Login/Registration, Browse/Search Marketplace, Listing Detail, Create/Edit Listing, In-App Chat, Rental Dashboard, Wishlist & Saved Searches, Sustainability Dashboard, and the Admin Dashboard (listing approval, user management, disputes, analytics). Section 6 provides initial-draft wireframes for the highest-priority screens.

5.2 Software / API Interfaces

Interface	Description
Frontend ↔ Backend REST API	React client communicates with the Node.js/Express backend over a versioned REST API, documented with an OpenAPI specification shared across module teams.
Backend ↔ AI Service	The Express backend calls the Python FastAPI AI service for listing suggestions, price estimation, semantic matching, and OCR duplicate detection.
Redis Pub/Sub Event Bus	Match events, rental reminders, and moderation events are published to Redis and consumed by the notification pipeline, decoupling producers from delivery.
Socket.io Channel	Real-time bidirectional channel used for in-app chat and live notification delivery to connected clients.
Payment Gateway API	External payment provider API used for storing payment methods and processing transactions.
Cloud Storage API	Cloudinary/AWS S3 API used to store and retrieve listing images and uploaded documents.
Notification Delivery API	Nodemailer (SMTP) and optional FCM push API used to reach users who are not actively connected.

5.3 Hardware Interfaces

- No dedicated hardware is required; the platform runs in a standard web browser on any desktop, laptop, tablet, or mobile device.
- Device camera or file picker is used to capture/upload item photos and QR codes for AI listing assistance and handover verification.

5.4 Communication Interfaces

- HTTPS/TLS for all REST API traffic between client and server.
- WebSocket (via Socket.io) for real-time chat and notification delivery.
- SMTP for outbound transactional and reminder emails.

6. Wireframes — UI/UX Mockups (Initial Draft)

The following low-fidelity wireframes represent the initial draft layout and interaction structure for CampusConnect's core screens. They are intended to validate information architecture and flow ahead of high-fidelity visual design (Phase 2, UI/UX Design).

6.1 Login / Registration

University-email-gated authentication is the entry point to the platform, satisfying FR-1.1 and FR-1.2.

CampusConnect Login

Welcome to CampusConnect
Sign in with your university email

University Email
student@university.edu

Password
XXXXXXXXXX

Sign In

[New here? Create an account](#)
Verified with university email domain only

Figure 2: Login / registration screen.

6.2 Browse Marketplace

Search, category filter chips, and a sidebar of advanced filters support FR-2.4, with a shortcut into AI-assisted listing creation.

The screenshot displays the 'Browse Marketplace' interface. At the top, a dark blue header contains the 'CampusConnect' logo on the left and a 'Browse Marketplace' link on the right. Below the header is a search bar with the placeholder text 'Search items, e.g. "calculus textbook"', a blue 'Search' button, and a blue '+ New Listing' button. Under the search bar are category filter chips: 'All', 'Textbooks', 'Electronics', 'Furniture', 'Rent', and 'Giveaway'. On the left side, there is a 'Filters' sidebar with sections for 'Category', 'Price Range', 'Condition', 'Distance', and 'Listing Type', each followed by an input field. The main content area features a grid of six placeholder listings. Each listing consists of a square image placeholder labeled 'photo', a price tag of '\$25 / week', and a blue 'Rent' button.

Figure 3: Browse / marketplace home screen.

6.3 Create Listing (AI-Assisted)

Photo upload triggers AI suggestions for category, condition, and price (FR-2.5, FR-3.1, FR-3.2), alongside an optional OCR duplicate check for notes/PYQs (FR-3.4).

CampusConnect

Create Listing

New Listing

Drag photos here or click to upload

🌟 AI Suggestion

Category, condition & price auto-filled from photo

Title

Category

Condition

Listing Type (Sell/Rent/Exchange/Giveaway)

Estimated Price (AI-assisted)

Description

Run OCR duplicate check (notes/PYQs)

Save Draft

Publish Listing

Figure 4: Create listing screen with AI-assisted suggestions.

6.4 Listing Detail & In-App Chat

Combines listing details, purchase/rental actions, and a real-time chat panel with the counterparty (FR-4.4).

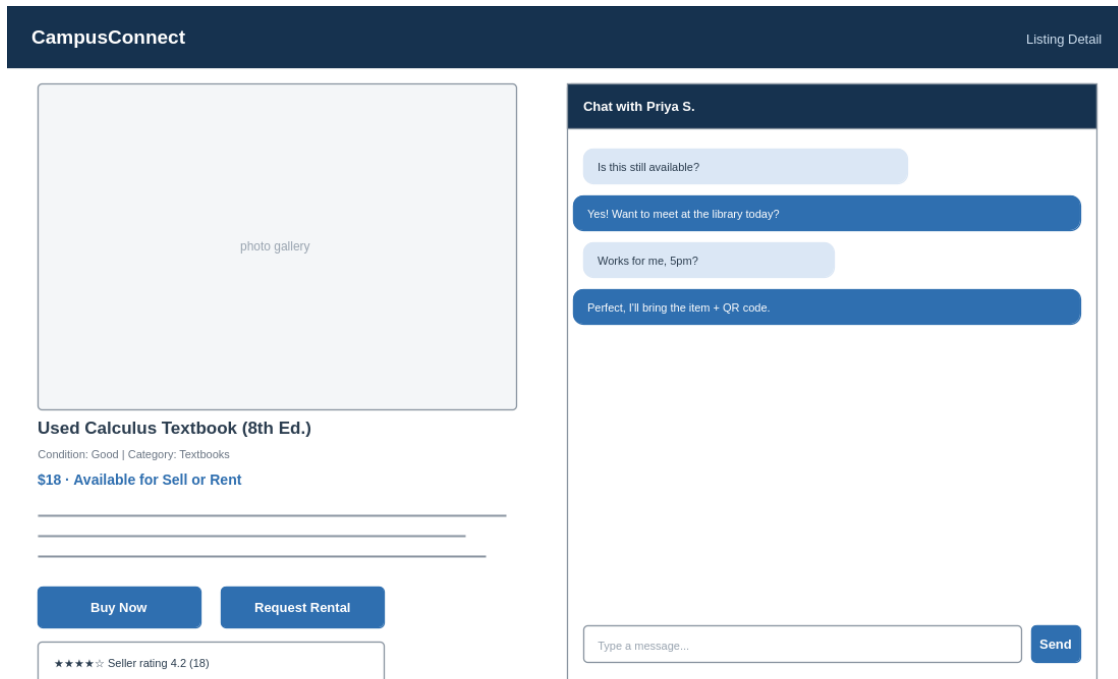


Figure 5: Listing detail with in-app chat panel.

6.5 Rental Dashboard

Gives renters and lenders a unified view of active rentals, requests, and history, along with due-date status and QR handover access (FR-5.3, FR-5.4, FR-5.5).

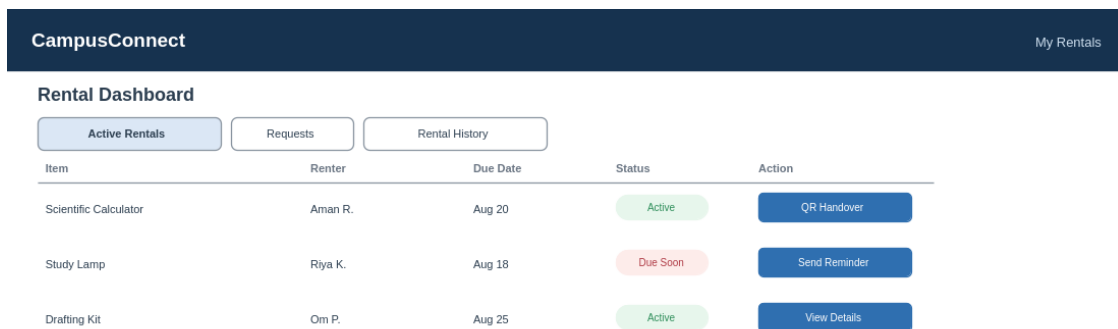


Figure 6: Rental dashboard.

6.6 Admin Dashboard

Provides administrators with at-a-glance metrics and a listing-approval queue supporting FR-6.1 and FR-6.5.

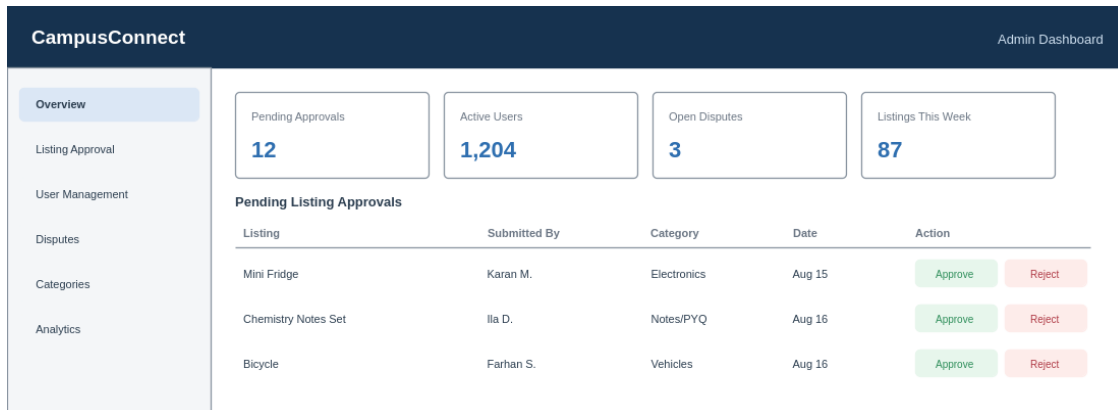


Figure 7: Admin dashboard — overview and listing approval.

7. Acceptance Criteria

Acceptance criteria are expressed in Given/When/Then form for the highest-priority user stories, defining the conditions under which each feature will be considered complete for QA sign-off.

AC-1 — University Email Registration (US-01)

- Given a prospective user submits a non-university email, when they attempt to register, then the system rejects the registration with a clear error message.
- Given a prospective user submits a valid university email, when they complete verification, then an account is created with the Student role.

AC-2 — AI-Assisted Listing Creation (US-03)

- Given a student uploads a photo during listing creation, when the AI service responds, then category, condition, and price fields are pre-filled and remain editable.
- Given the AI service is unavailable or times out, when the student continues, then the listing form remains usable for fully manual entry without an error blocking submission.

AC-3 — Wanted Post Matching (US-05)

- Given a student has an open wanted post, when a new listing is published that semantically matches it, then a match notification is delivered to the student within the expected notification latency.
- Given no matching listing exists, when time passes, then no false-positive match notification is sent.

AC-4 — In-App Chat (US-06)

- Given two users are both online, when one sends a message, then the other receives it in real time without a page refresh.
- Given a recipient is offline, when a message is sent, then the message is stored and displayed the next time the recipient opens the chat.

AC-5 — QR Code Handover (US-08)

- Given a rental has been approved, when the lender generates a handover QR code, then the code uniquely identifies that specific transaction.
- Given the borrower scans a valid QR code at pickup, when verification succeeds, then the transaction state updates to Active.
- Given an invalid or already-used QR code is scanned, when verification is attempted, then the system rejects it and displays an explanatory error.

AC-6 — Rental Due-Date Reminders (US-09)

- Given a rental has a due date, when the reminder window is reached, then a reminder notification is sent to the borrower automatically without manual triggering.
- Given an item is returned before the due date, when the return is confirmed, then no further reminders are sent for that rental.

AC-7 — Listing Approval Workflow (US-15)

- Given a student submits a new listing, when moderation is enabled for its category, then the listing is held in a Pending state until an administrator approves or rejects it.

- Given an administrator rejects a listing, when the action is submitted, then the submitting student receives a moderation notification with the reason.

AC-8 — Payment and Receipt Confirmation (US-19)

- Given a buyer completes payment, when the seller has not yet confirmed item handover, then transaction funds remain in a pending/held state.
- Given both buyer and seller confirm receipt independently, when both confirmations are recorded, then the transaction is marked Completed.

AC-9 — OCR Duplicate Detection (US-10)

- Given a student uploads notes/PYQ files that closely match an existing listing, when the OCR check runs, then the system flags the potential duplicate before publication.
- Given uploaded content has no meaningful overlap with existing listings, when the OCR check runs, then the listing is allowed to publish without a duplicate flag.

8. Appendix

8.1 Requirements Traceability — Module Ownership

Maps each functional requirement group to its accountable owner, per the Project Plan's team task assignments.

FR Group	Module	Owner
FR-1.x	Authentication & User Management	Dhrumil Doshi
FR-2.x	Marketplace & Listing Management	Nishka Shah
FR-3.x	AI & Intelligent Services	Heer Mehta
FR-4.x	Search, Chat & Notification	Heer Shah
FR-5.x	Rental & Transaction Management	Charmi Bhayani
FR-6.x	Admin Panel & Analytics	Kajal Varlani
FR-7.x	Platform / Cross-Cutting (Architecture, DevOps)	Dev Barot, Deep Boghara

8.2 Assumptions

- Every user has a valid, single university email account used for identity verification.
- The AI and matching services are treated as internal microservices; no third-party AI vendor lock-in beyond the Groq API is assumed at this stage.
- Payment processing relies on a third-party payment gateway; CampusConnect does not store raw card data directly.

8.3 Open Questions for Design Review

- Should listing moderation (FR-6.1) apply to all categories, or only flagged/high-risk categories, to avoid slowing down low-risk listings?
- What is the acceptable false-positive rate for OCR duplicate detection before it becomes a usability problem?
- Should the sustainability dashboard's CO₂ impact estimate be sourced from a fixed heuristic or a configurable model?